Xinyan Wang

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Research Interests

LLM post-training (RL, on-policy distillation), LLM agents and agent red-teaming, efficient inference.

Education

- **University of Wisconsin–Madison** Madison, WI, USA
  Ph.D. in Statistics; Advisors: Profs. Jun Shao and Xiao Luo *Sep 2023 – May 2028 (Expected)*
  M.S. in Computer Science *Sep 2023 – Present*
  M.S. in Statistics *Sep 2022 – May 2023*
  Visiting Student in Statistics and Computer Science *Sep 2021 – May 2022*
- **East China Normal University** Shanghai, China
  B.S. in Statistics *Sep 2018 – June 2022*

Experience

- **SaFo Lab** Johns Hopkins University
  Research Intern; Mentor: Prof. Chaowei Xiao *Aug 2025 – Present*
  - Research on safe and efficient large reasoning models and LLM agents.

Publications

Xinyan Wang*, Xiaogeng Liu*, Jun Shao, Chaowei Xiao. *Appreciation Teaches Unauthorized Execution: Self-Evolution Degrades the Robustness of Personal Agents to Indirect Prompt Injection.* Preprint, 2026. arXiv:2610.39122.

Xiaogeng Liu, **Xinyan Wang**, Chaowei Xiao. *FlipJev: Fast Black-Box Red Teaming Finds Universal Adversarial Suffixes for Typed Decision Models.* Preprint, 2026. arXiv:2610.38895.

Xiaogeng Liu, **Xinyan Wang**, Yingzi Ma, Yechao Zhang, Chaowei Xiao. *When Are Teacher Tokens Reliable? Position-Weighted On-Policy Self-Distillation for Reasoning.* NeurIPS 2026 (Poster). arXiv:2605.21606.

Xiaogeng Liu, **Xinyan Wang**, Yechao Zhang, Sanjay Kariyappa, Chong Xiang, Muhao Chen, G. Edward Suh, Chaowei Xiao. *ReasoningBomb: A Stealthy Denial-of-Service Attack by Inducing Pathologically Long Reasoning in Large Reasoning Models.* ACM Conference on Computer and Communications Security (CCS), 2026.

Xinyan Wang, Xiaogeng Liu, Ming Pei, Chaowei Xiao. *ROM: Real-time Overthinking Mitigation via Streaming Detection and Intervention.* Preprint, 2026. arXiv:2603.22016.

Jun Shao, **Xinyan Wang**. *MLE with Datasets from Populations Having Shared Parameters.* Statistical Theory and Related Fields, 7(3):213–222, 2023.

Research Experience

- **Evolve-IPI: Self-Evolution and Indirect Prompt Injection** JHU
  With Prof. Chaowei Xiao (Preprint, 2026)
  - Studied personal agents across five tasks, separating adaptation with authorized user feedback from tests with unauthorized instructions embedded in external content.
  - Found that self-evolution raises average attack success by 14.0–48.8 percentage points under skill evolution and 3.0–39.3 points under reinforcement learning compared with base agents.
- **FlipJev: Red-Teaming Typed Decision Models** JHU
  With Prof. Chaowei Xiao et al. (Preprint, 2026)
  - Studied fast black-box red teaming of typed decision models using an evolutionary suffix search guided by output probabilities, without an attacker language model.
  - Found universal suffixes that flip 43% of unseen documents; a suffix learned on one task flips 90–100% of samples across five other tasks without further search.

- **PW-OPSD: Position-Weighted On-Policy Self-Distillation for Reasoning** JHU
  With Prof. Chaowei Xiao et al. (NeurIPS 2026)
  - Built a branch-viability diagnostic that forces privileged-teacher token alternatives after the student's on-policy prefix and tests whether the correct answer still emerges.
  - Showed on Qwen3-4B that within-sequence position predicts teacher-token reliability far better than local uncertainty (AUROC 0.83 vs. 0.57); the resulting PW-OPSD up-weights later tokens, improving AIME 2024/2025 Avg@12 by +1.0/+1.1 at no extra teacher compute.

- **ReasoningBomb: A Denial-of-Service Attack on Large Reasoning Models** JHU
  With Prof. Chaowei Xiao et al. (ACM CCS 2026)
  - Formalized prompt-induced inference-time denial-of-service (PI-DoS) for LRMs and built a reinforcement-learning attacker that crafts short, natural prompts inducing pathologically long reasoning.
  - Across 7 open-source and 3 commercial models, induced 286.7× input-to-output amplification and bypassed input-, output-, and dual-stage detectors at 99.8%, 98.7%, and 98.4%.

- **ROM: Real-time Overthinking Mitigation in Large Reasoning Models** JHU
  First author; advised by Prof. Chaowei Xiao
  - Showed overthinking in large reasoning models is a latent, decodable productive-to-redundant transition, and built ROM, a streaming detector ($\sim$0.1% of parameters) intervening on a frozen LRM in real time.
  - Attained the highest accuracy in 19 of 25 settings across five backbones and five benchmarks, cutting response length 28–77% and latency 46.5%.

- **MLE with Datasets from Populations Having Shared Parameters** UW–Madison
  Advisor: Prof. Jun Shao (Statistical Theory and Related Fields, 2023)
  - Studied maximum likelihood estimation with multiple datasets from heterogeneous populations sharing parameters; established asymptotic normality, estimated variability via bootstrap, and reduced the estimator's standard deviation by nearly 57% versus single-dataset MLEs.

Teaching

STAT 240 (Data Science Modeling I), STAT 309 (Introduction to Probability and Mathematical Statistics I), STAT 340 (Data Science Modeling II), STAT 609 (Introduction to Statistical Inference), STAT 610 (Statistical Inference), STAT 611 (Statistical Models for Data Science), STAT 613 (Statistical Methods for Data Science).

Professional Service

Reviewer: ACL 2026, ECCV 2026, EMNLP 2026, NeurIPS 2026